# WEEK 8: Student Performance Dashboard

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## 1. Executive Summary

This report presents the analysis and visualization of the StudentsPerformance dataset using Microsoft Power BI. The objective is to evaluate student achievement metrics, understand the impact of parental education and test preparation courses, and track overall pass/fail distributions.

## 2. Dataset Overview & Data Modeling

- **Source Data:** The StudentsPerformance dataset including metrics for average scores, grades, pass/fail status, parental education levels, test preparation status, lunch plans, and gender.
- **Data Transformation:** Performed data cleaning, type conversions, and generated necessary conditional columns using Power Query.
- **Substitutions:** Utilized the gender field as a proxy count metric for aggregation in visual elements.

## 3. Key Performance Indicators (KPI Cards)

The dashboard incorporates high-level summary KPI cards to display critical metrics at a glance:

- Pass%: 0.97 (97\% overall pass rate)
- Average Reading Score: 69.17
- Average Writing Score: 68.05
- Average Math Score: 66.09
- Total Marks / Count Metrics: Evaluated across student segments.

## 4. Visualizations & Analytical Breakdown

### 4.1 Clustered Bar Chart: Parental Level of Education vs. Scores

- Visual Type: Clustered Bar Chart.
- Insight: Compares average math, reading, and writing scores against different tiers of parental education (from master's degree down to high school).

### 4.2 Donut Charts: Grade & Pass/Fail Distribution

- Visual Type: Donut Charts.
- Insight:
- Displays the proportional breakdown of student grades (A, B, C, E, F).
- Highlights the high pass ratio (97\%) versus the fail ratio (3\%).

### 4.3 Pie Chart & Stacked Column Chart: Test Prep & Gender Analysis

- Visual Type: Pie Chart and Stacked Column Chart.
- Insight: Illustrates the participation rates in test preparation courses and compares performance metrics across genders relative to their test prep and lunch categories.

## 5. Interactivity & Slicers

The dashboard features dedicated slicers for Gender, Test Preparation Course, and Lunch Type, enabling dynamic filtering to analyze specific student subgroups in real-time.

# 6. Dashboard Visualization / Final Output

